Rahul Joseph Bejoy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.40
St. Thomas Central School, Mukkolakkal, Thiruvananthapuram, Kerala, India	Class XII (CBSE)	2024	99.00%
The School of The Good Shepherd, Akkulam, Thiruvananthapuram, Kerala, India	Class X (CBSE)	2022	99.40%

SCHOLASTIC ACHIEVEMENTS

Highest CGPA in the Chemical Engineering department after four semesters (CGPA: 9.40/10)

EXPERIENCE

CSIR - National Institute for Interdisciplinary Science and Technology (CSIR-NIIST) | *Research Intern, Advanced Functional Materials Division* | Thiruvananthapuram, Kerala | *May 2026 - June 2026*

- Conducted research on the "Preparation and Characterization of Polyether ether ketone (PEEK) Aerogels" under the guidance of Dr. Bhoje Gowd E.
- Utilised specialised thermal analysis software to evaluate characterisation metrics, identifying critical thermodynamic markers including glass transition temperature (Tg) and melting point (Tm).
- Processed and interpreted complex experimental datasets to evaluate the structural and thermal efficiency of synthesised aerogel samples.

RELEVANT COURSES

Chemical Core:

- Chemical Engineering Thermodynamics, Fluid Mechanics, Chemical Reaction Engineering, Heat Transfer, Material Science for Chemical Engineers, Mechanical Unit Operations

Japanese Courses:

- Japanese Culture and Society, Spoken Japanese Basics

Mathematics & Computation:

- Calculus(1 & 2), Differential Equations, Linear Algebra, Numerical Methods(Solving Linear & Non-linear Equations, Eigenvalue Problems, Numerical Integration, Initial & Boundary Value Problems, Partial Differential Equations), Applied Mathematics for Chemical Engineers(Probability Theory & Distributions, Hypothesis Testing & ANOVA, Regression Analysis, Maximum Likelihood Estimation, Linear Algebra & PCA)

SKILLS

Programming Languages: C/C++, Python, SQL,Java

Libraries & Tools: NumPy, Pandas, Matplotlib, Flask, React, Git

Core CS: Data Structures & Algorithms, OOP, Database Systems, Machine Learning, Numerical Methods

PROJECTS

LLM From Scratch -- GPT-Style Language Model Implementation | Github link

- Implemented a GPT-style large language model from scratch in PyTorch, covering tokenization, byte-pair encoding, multi-head self-attention, causal masking, and transformer blocks.
- Built custom tokenizers (word-level and BPE via tiktoken) and a sliding-window dataloader for next-token prediction training pairs.
- Designed and implemented causal (masked) self-attention and multi-head attention mechanisms with dropout regularization, verified against reference outputs at each stage.
- Assembled a full transformer-based GPT architecture (embedding layers, transformer blocks, layer normalization, residual connections) capable of autoregressive text generation.

Fake News Detector using NLP & TF-IDF | Github link

- Developed a fake news classification system using TF-IDF vectorization and Logistic Regression, achieving 99.29% accuracy on a dataset of 44,000+ news articles across Fake and Real categories.
- Engineered a text preprocessing pipeline including URL removal, lowercasing, and bigram feature extraction with 50,000 TF-IDF features, combining article title and body for richer signal representation.

POSITIONS OF RESPONSIBILITY

Internship Coordinator | Office of Career Services

- Actively pitched internship opportunities to prospective companies, building and expanding partnership networks.
- Screened and verified student CVs to ensure eligibility and quality for internship applications.
- Will serve as the primary point of contact for partner companies throughout the internship cycle, coordinating logistics and resolving queries.

Core | Quiz Club

- Managed quiz operations including scoring, verification and event logistics.